Neutral-Atom Quantum Computing Architecture for Cryptographic Simulation.

Neutral-atom quantum processors exploit optical tweezer arrays to trap individual rubidium and ytterbium atoms within ultra-high vacuum chambers. Two-qubit gate operations are mediated via laser excitation to high-lying Rydberg states, providing strong, switchable dipole-dipole interactions with high physical fidelity.

Our research benchmarks surface code lattice error thresholds against coherent laser phase noise and spontaneous atomic emission events across a two-dimensional grid of one hundred physical qubits.